4 Northern Bank has two types of account.
Both accounts pay compound interest.

Cash savings account	**Shares account**
Interest	Interest
2.5% per annum	3.5% per annum

Ali invests £2000 in the cash savings account.
Ben invests £1600 in the shares account.

(a) Work out who will get the most interest by the end of 3 years.
You must show all your working.

Ali: 100% + 2.5% = 102.5%, so multiplier 1.025

After 1 year: $2000 \times 1.025 = £2050$

1 mark
Apply one year of interest

After 3 years: $2000 \times 1.025^3 = £2153.78125$

Interest = 2153.78125 – 2000 = £153.78 (2 d.p.)

1 mark
Compound over three years

Ben: multiplier = 1.035

After 3 years: $1600 \times 1.035^3 = £1773.9486$

Interest = 1773.9486 – 1600 = £173.95 (2 d.p.)

1 mark
Complete both interest calculations

1 mark
Ben earns more interest

Ben: £173.95 > £153.78. **(4)**

In the 3rd year the rate of interest for the shares account is changed to 4% per annum.

(b) Does this affect who will get the most interest by the end of 3 years?
Give a reason for your answer.

No. Ben already earns more interest; raising his final-year rate to 4% increases his interest further, so he still wins.

1 mark
Same conclusion, supported by the increased rate **(1)**

(Total for Question 4 is 5 marks)

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